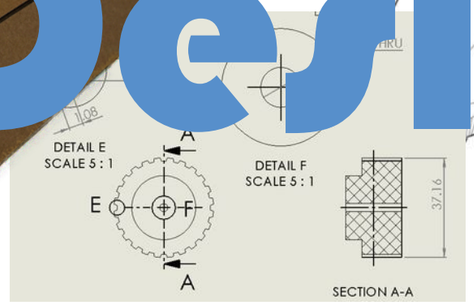
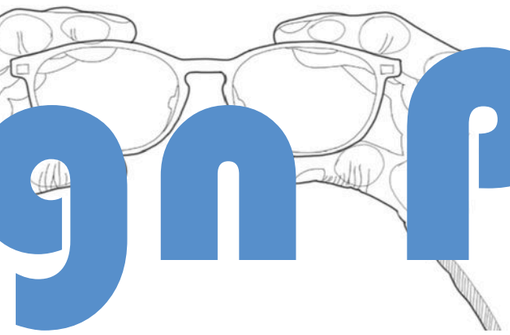
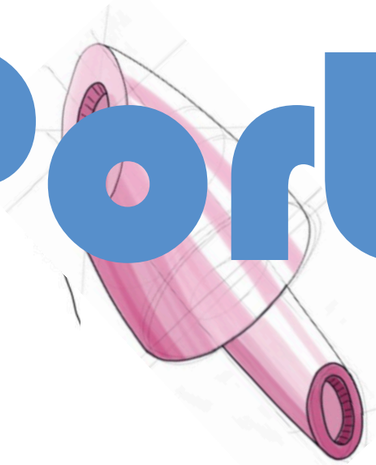
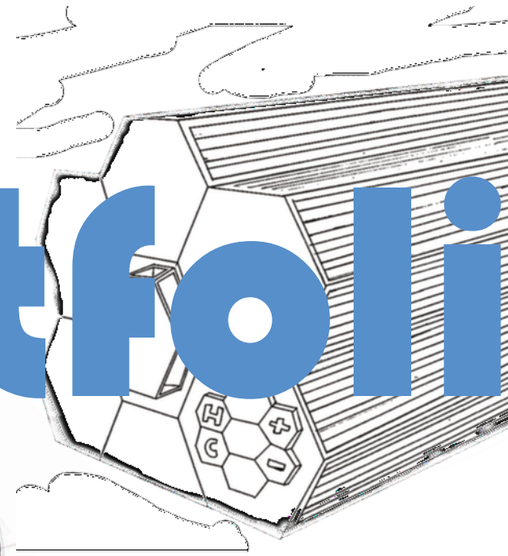
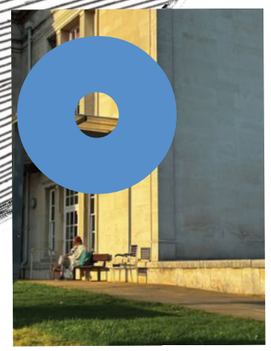












Design Portfolio

Rachel Zinkin

Drawing Pages 3-7

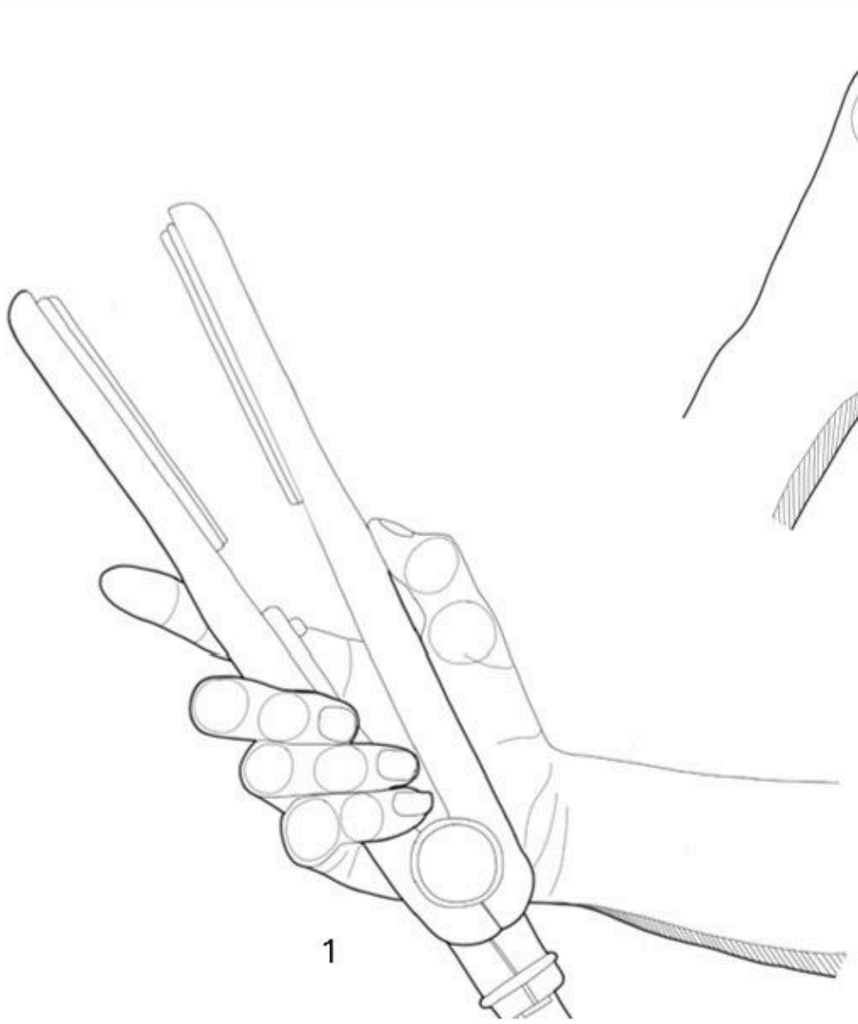
Project work Pages 8-21

Technical skills Pages 22-26

Drawing skills

Drawing - digital drawing: human interface assignment

4

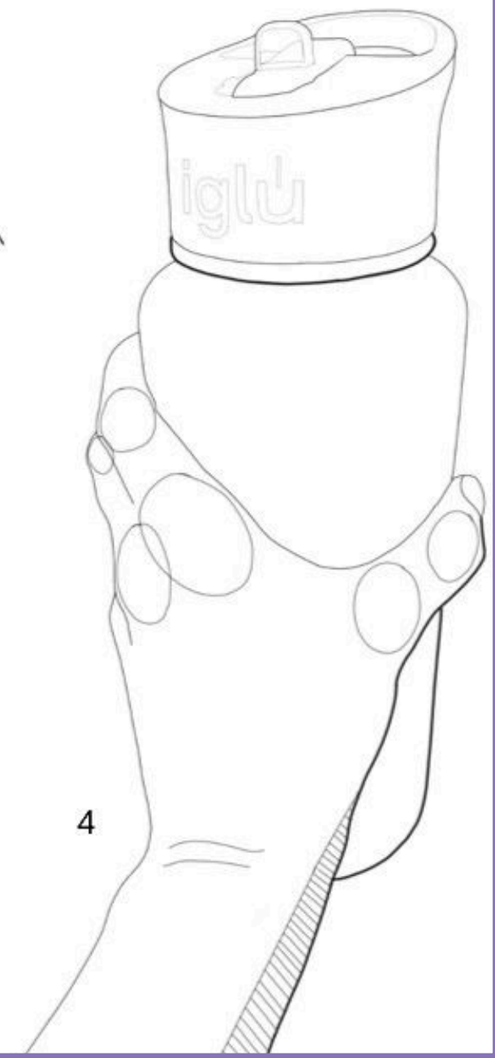


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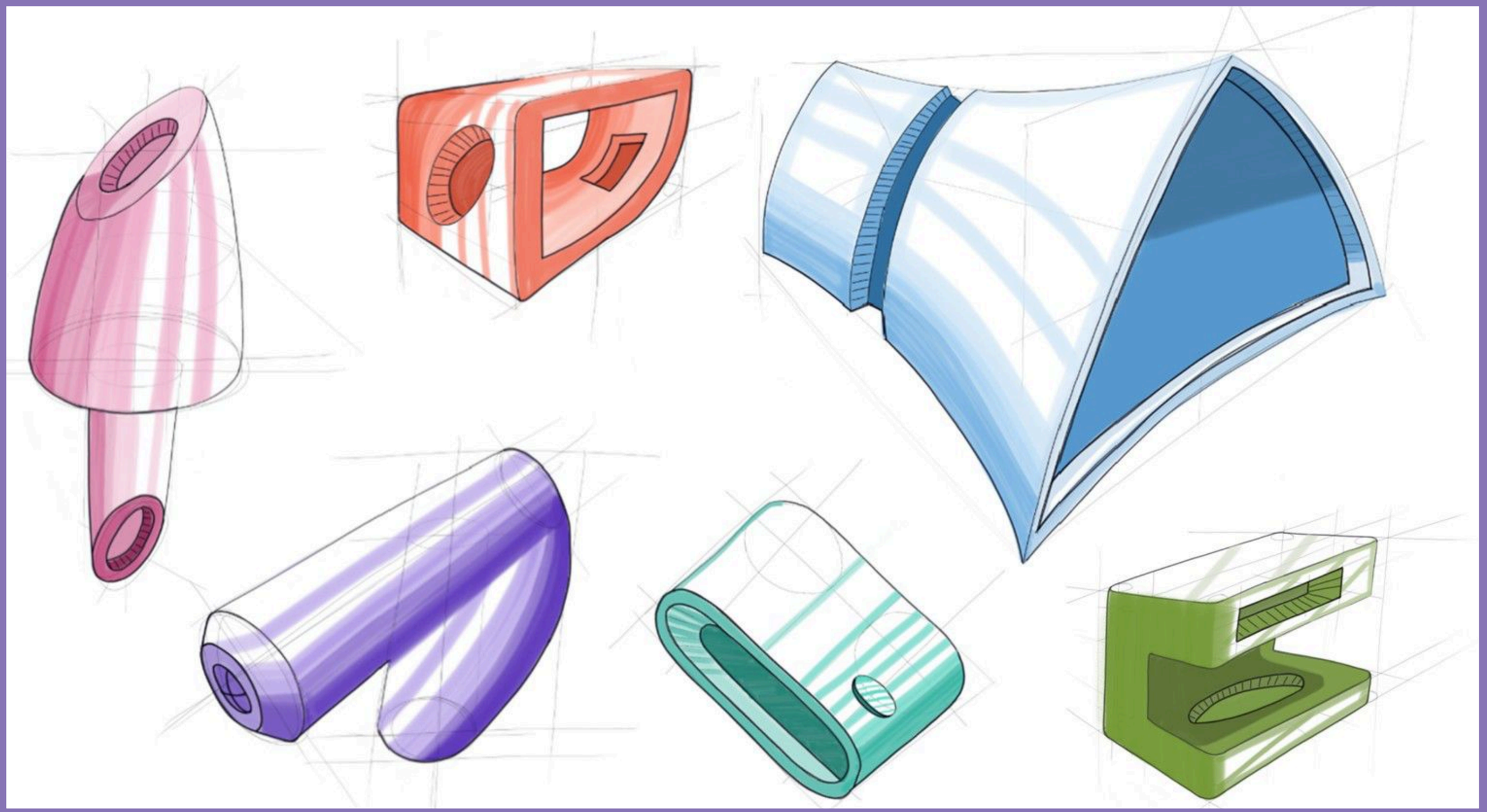
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3



4

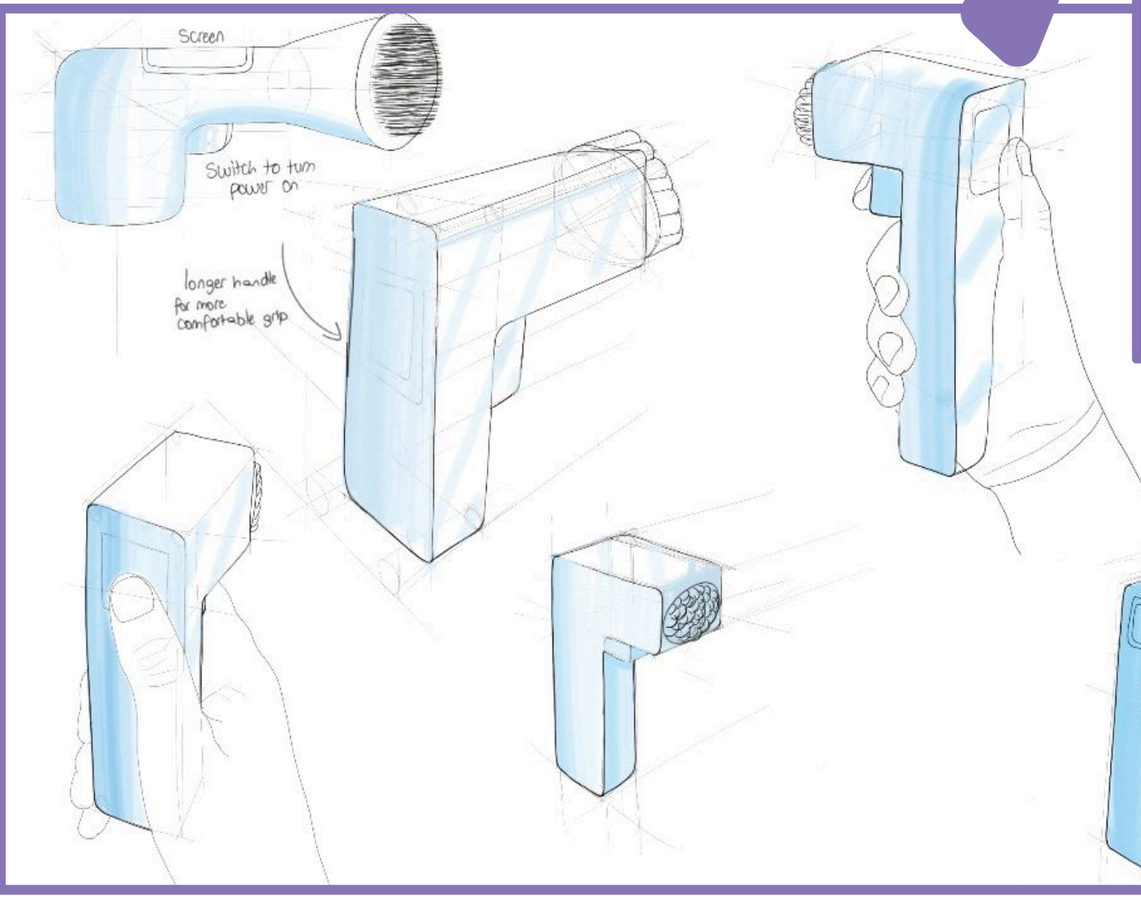
Digital drawing assignment to explore ergonomics of products and human interface.

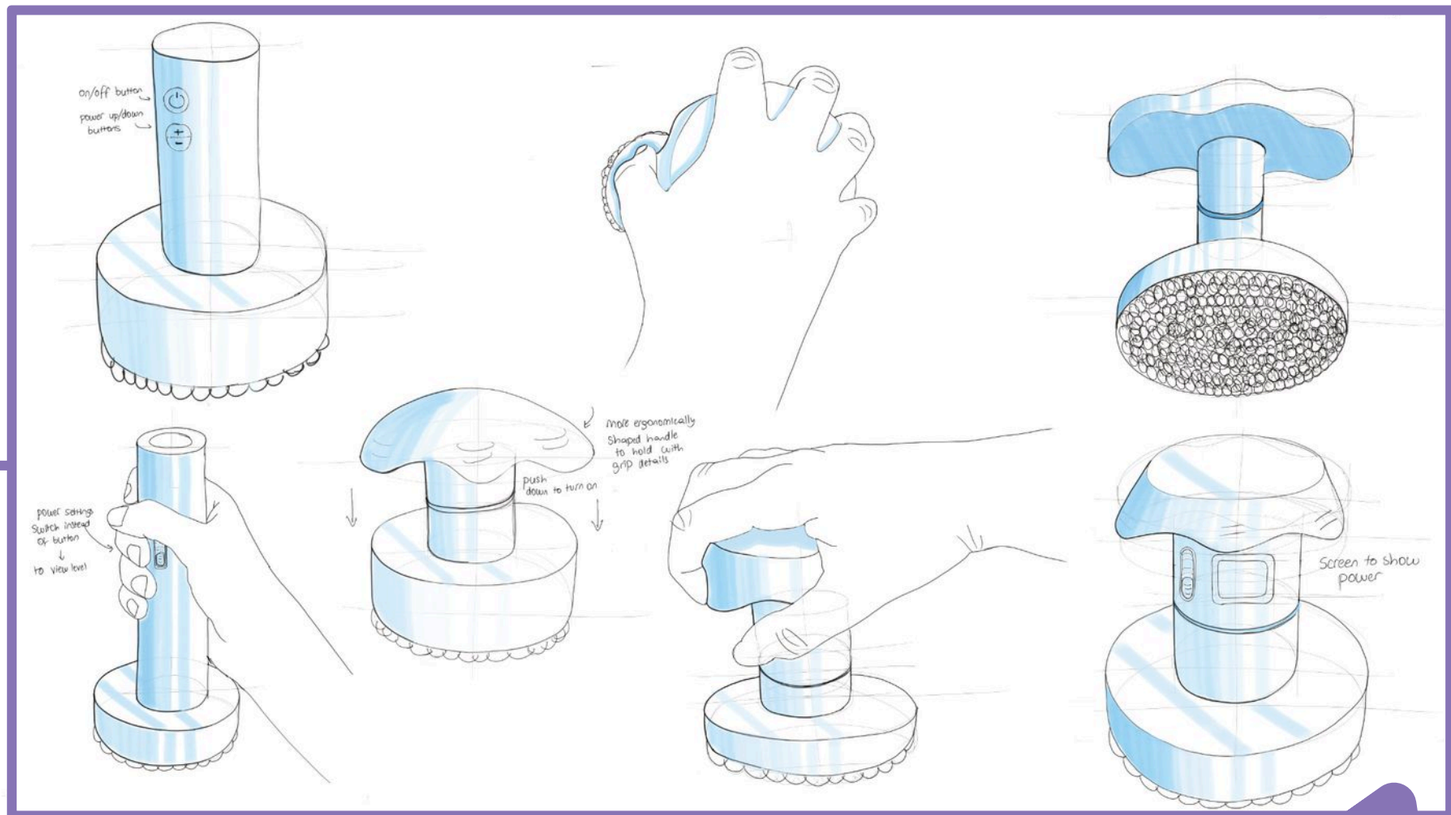


Digital drawing using marker to depict light and shade.

Drawing - digital drawing: concept development of scrub brush 6

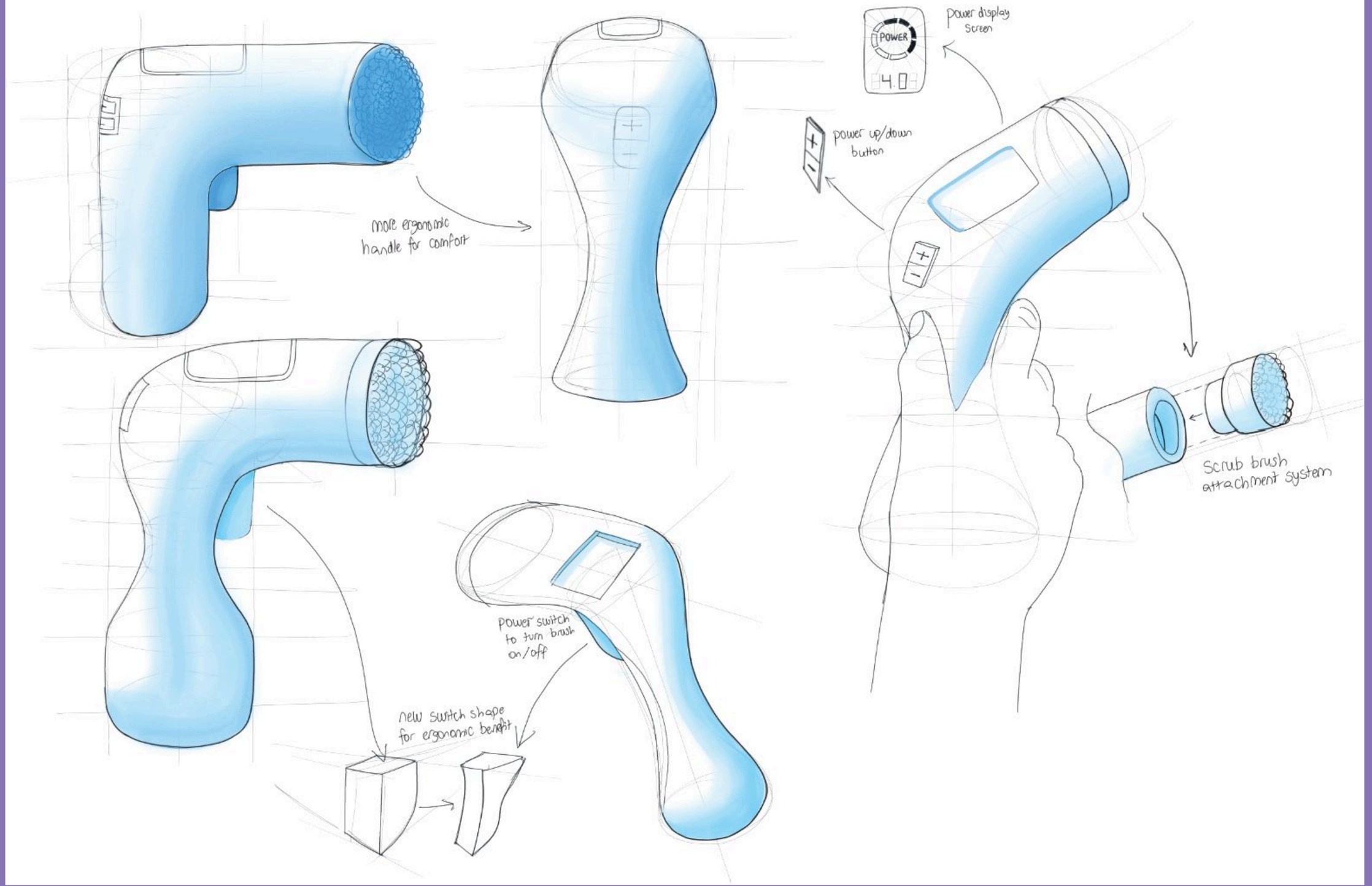
Concept development 1





Concept development 2

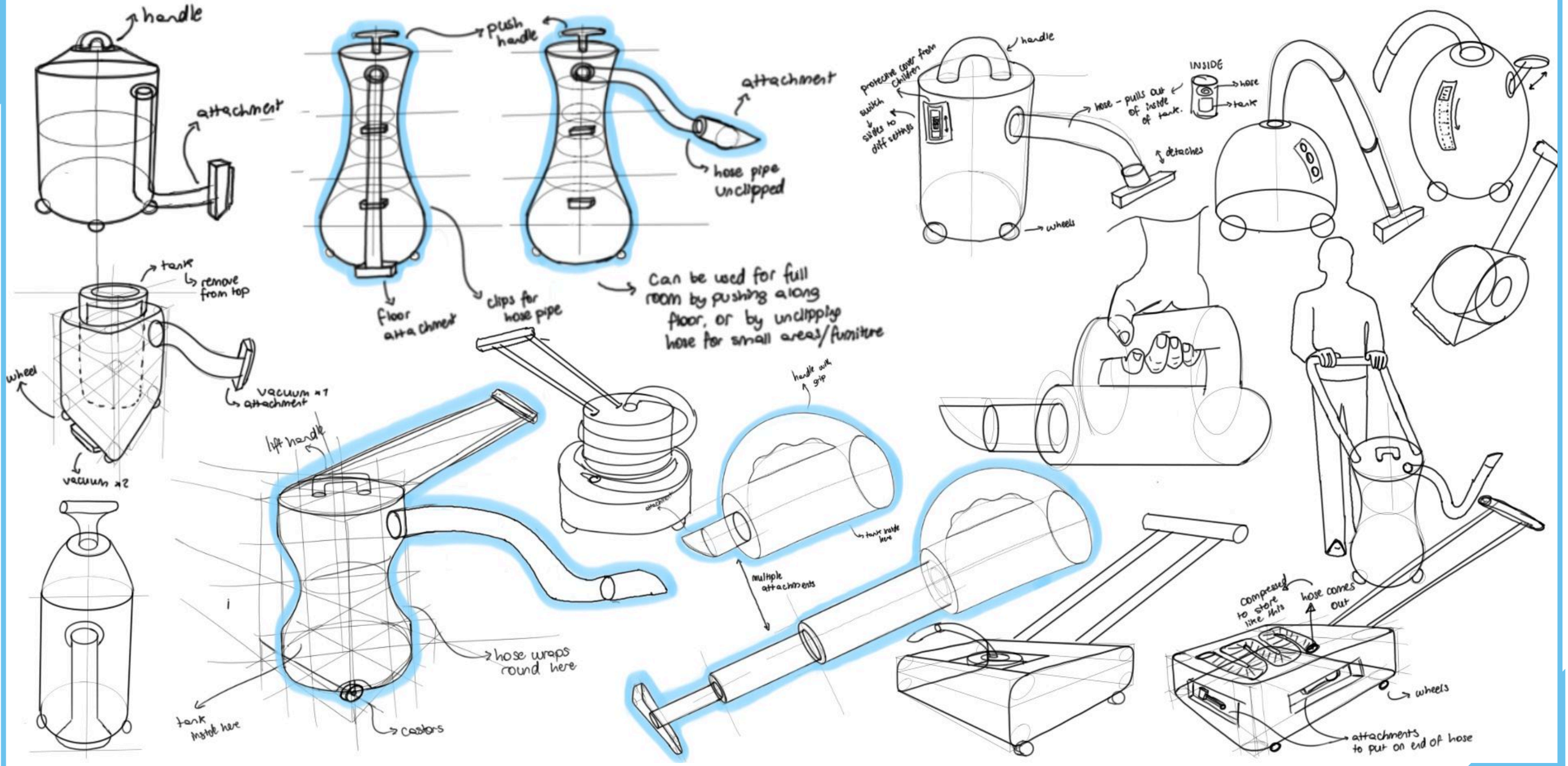
Drawing - digital drawing: refinement of scrub brush design



Project work

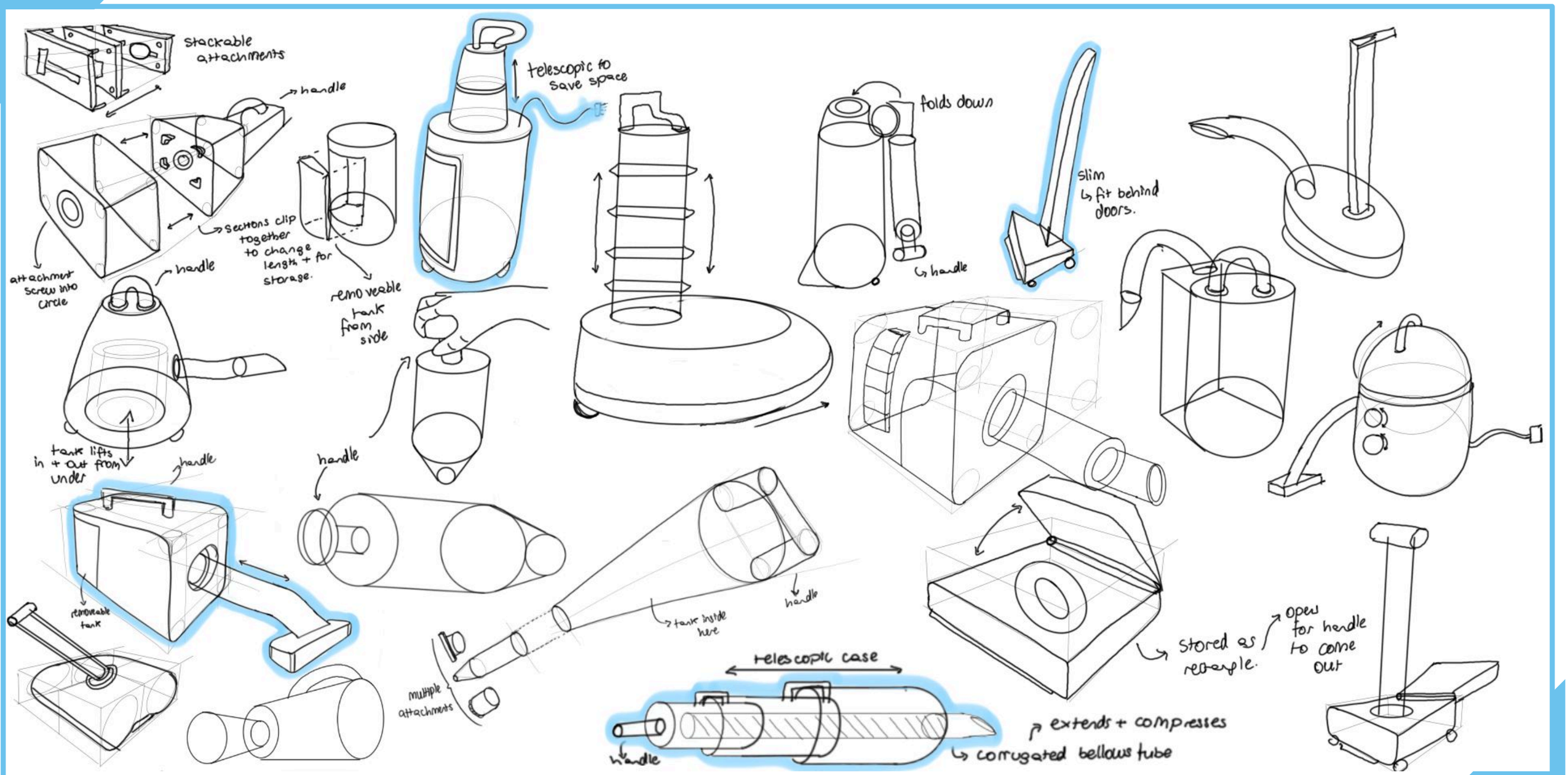
Wet and dry vacuum

Project work: wet and dry vacuum for daycare - ideation



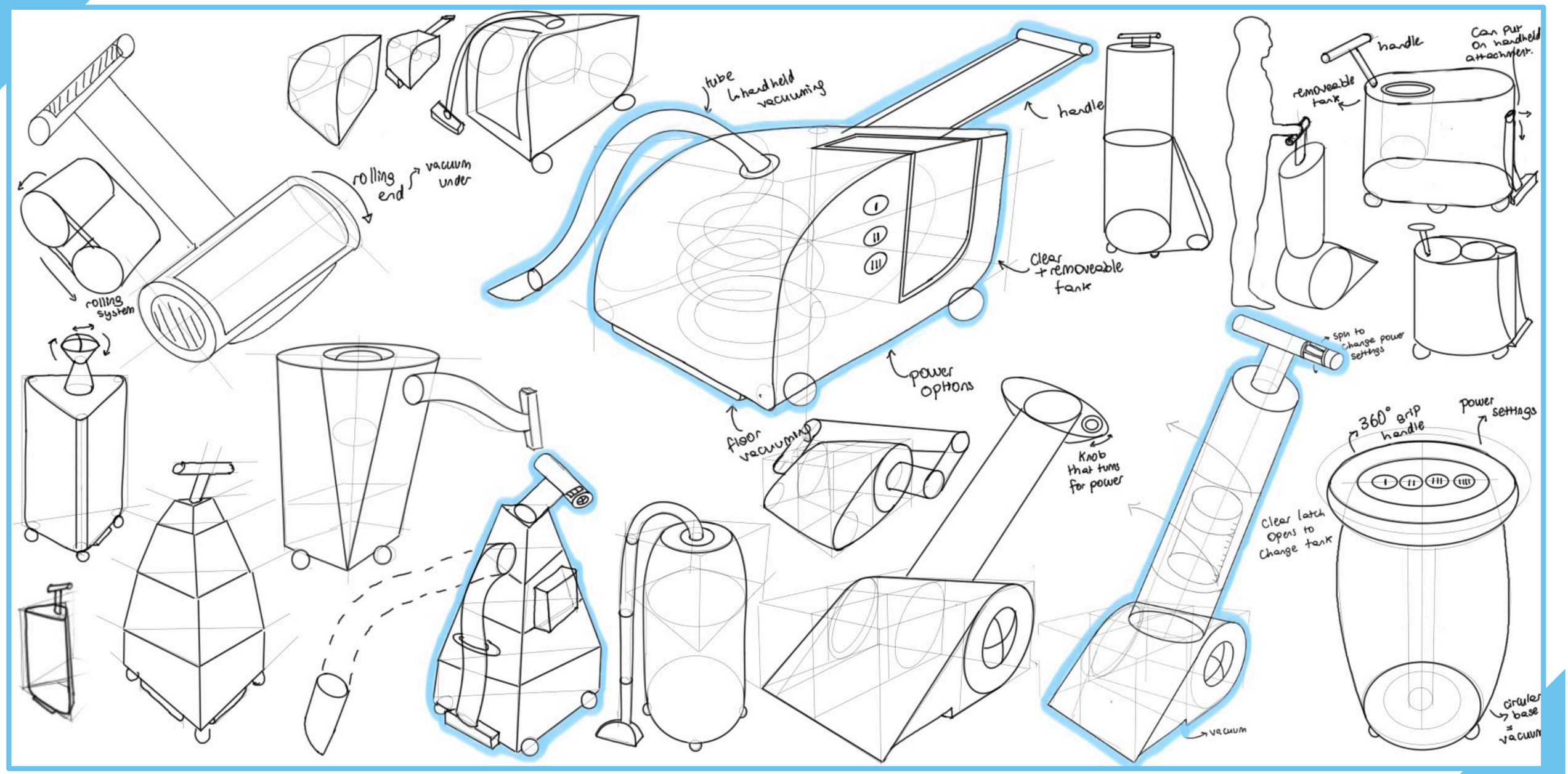
Digital sketching for early stages of ideation for the target market of use in a 'daycare'. The most successful ideas in my opinion are highlighted.

Project work: wet and dry vacuum for students - ideation



Digital sketching for early stages of ideation for the target market of use by 'students'. The most successful ideas in my opinion are highlighted.

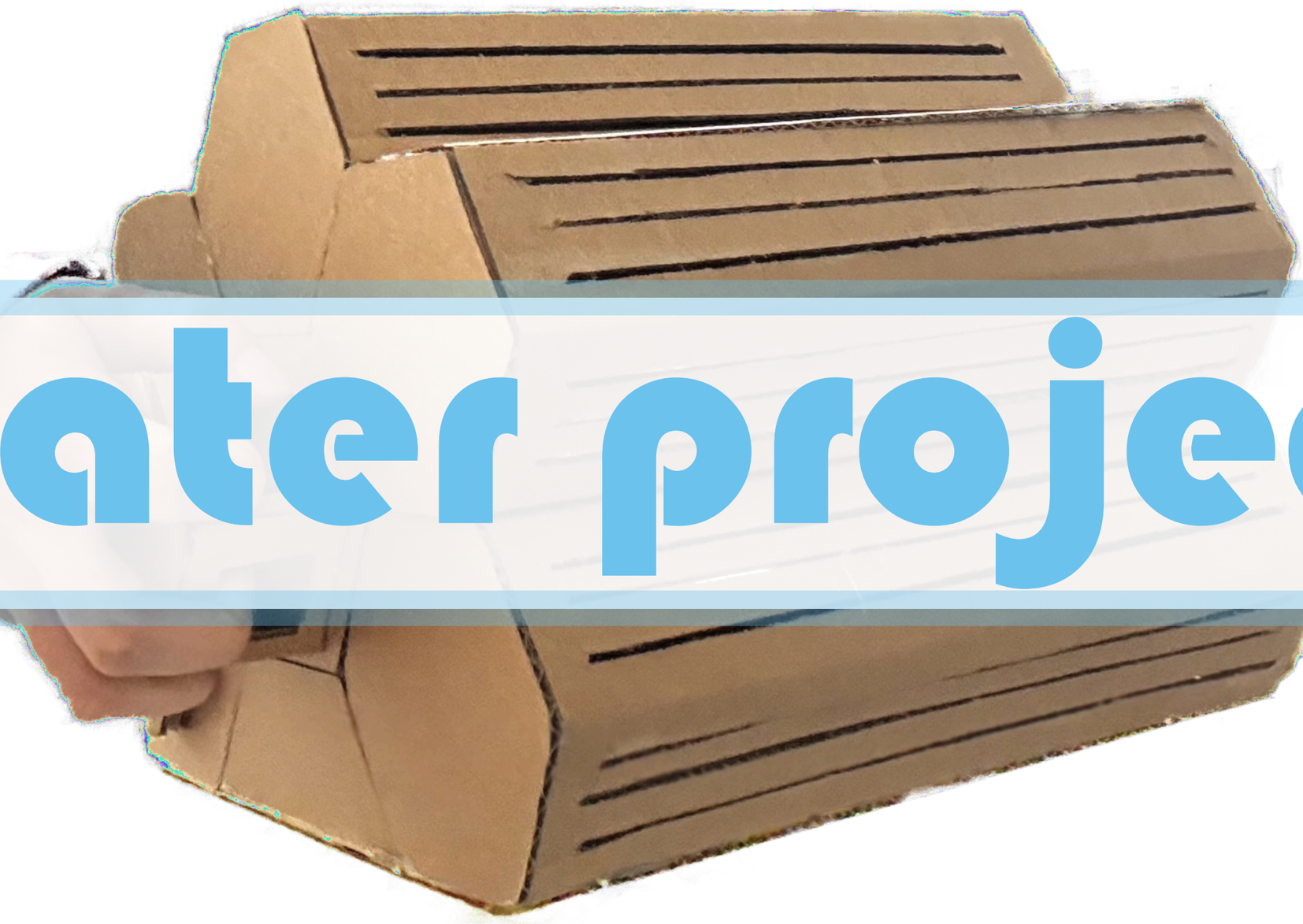
Project work: wet and dry vacuum for care homes - ideation



Digital sketching for early stages of ideation for the target market of use in a 'care home'. The most successful ideas in my opinion are highlighted.

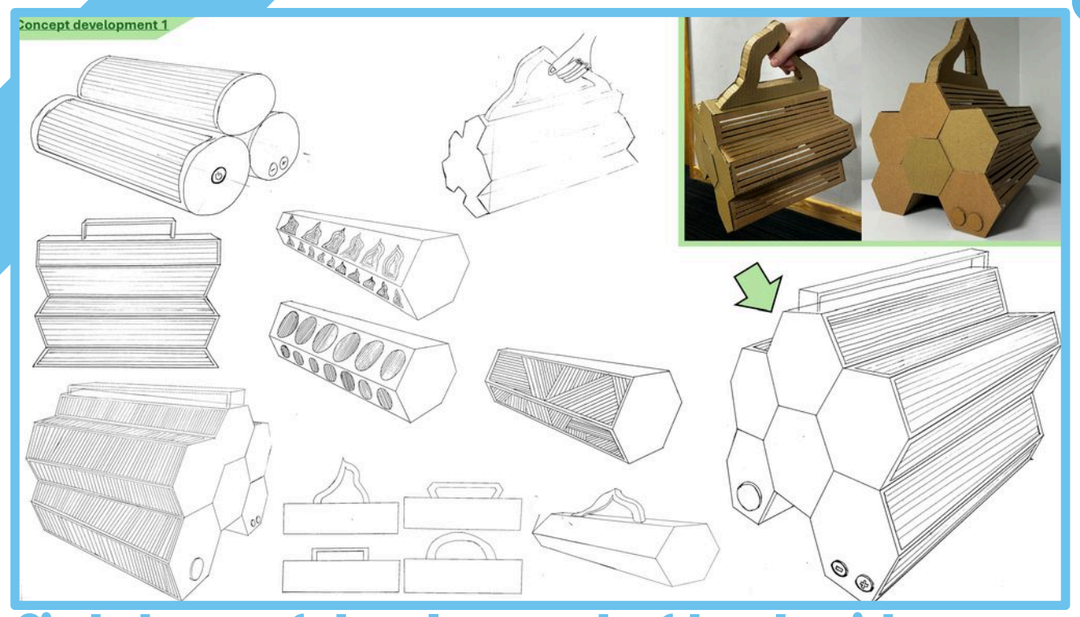
Upload pending

By October 21st

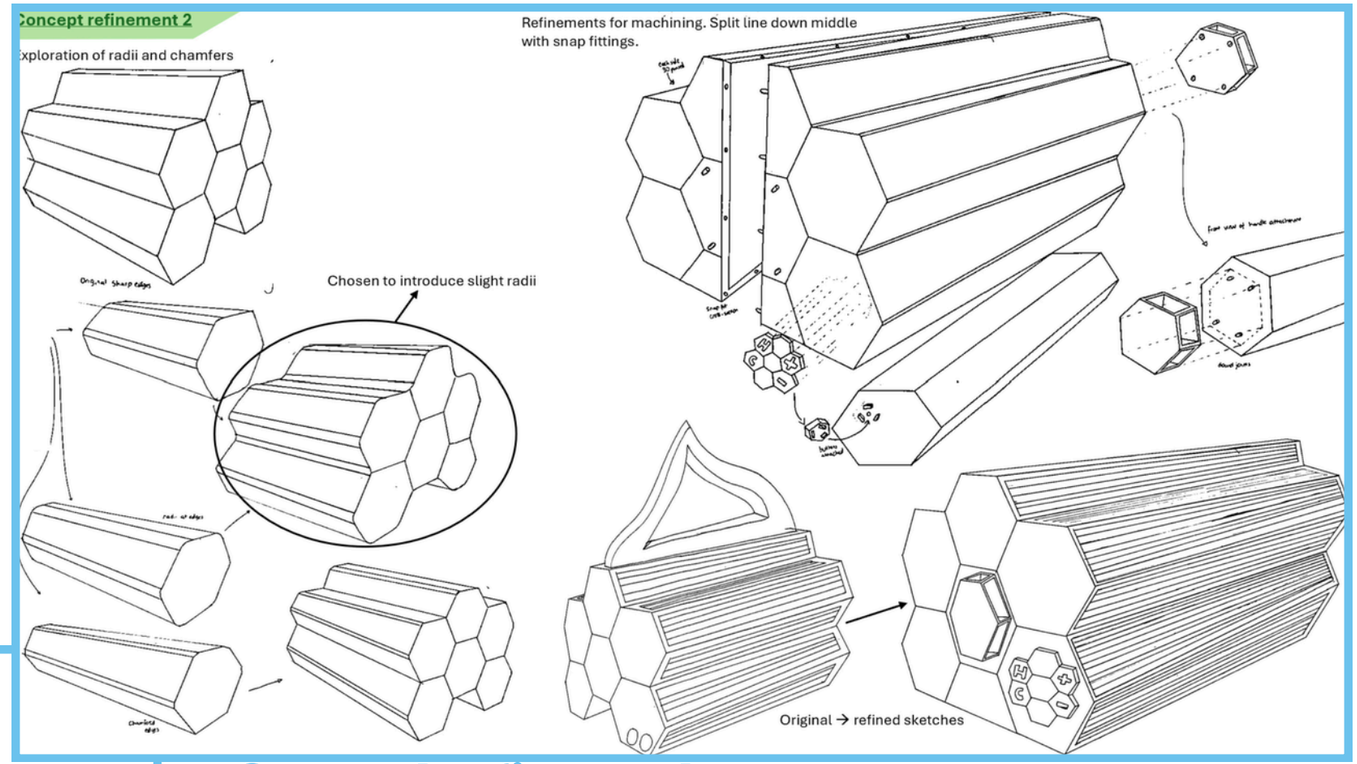


Heater project

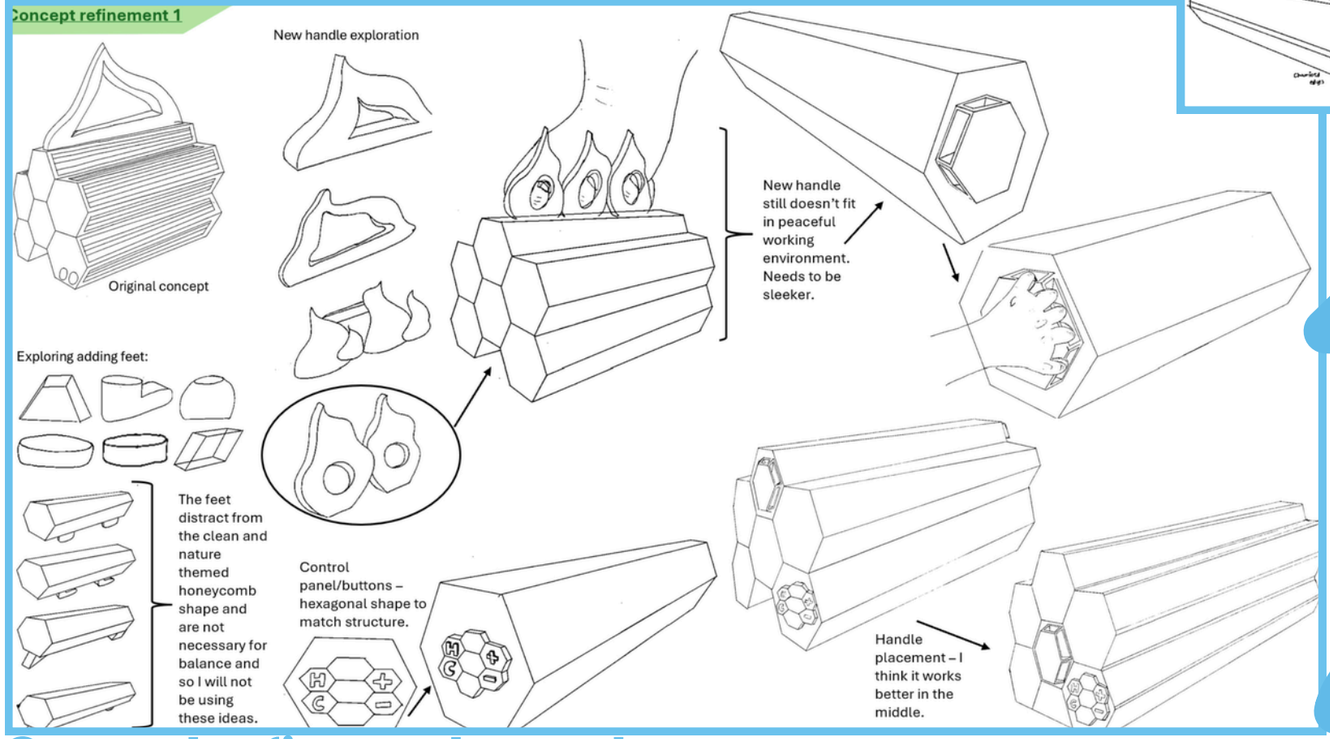
Project work: Heater project - ideation + refinement drawing



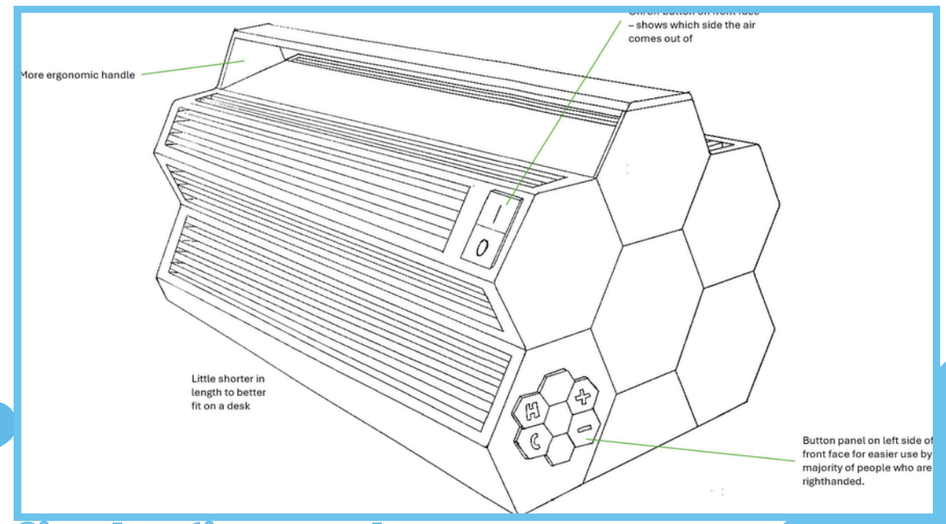
First stage of development of heater idea



Concept refinement page one

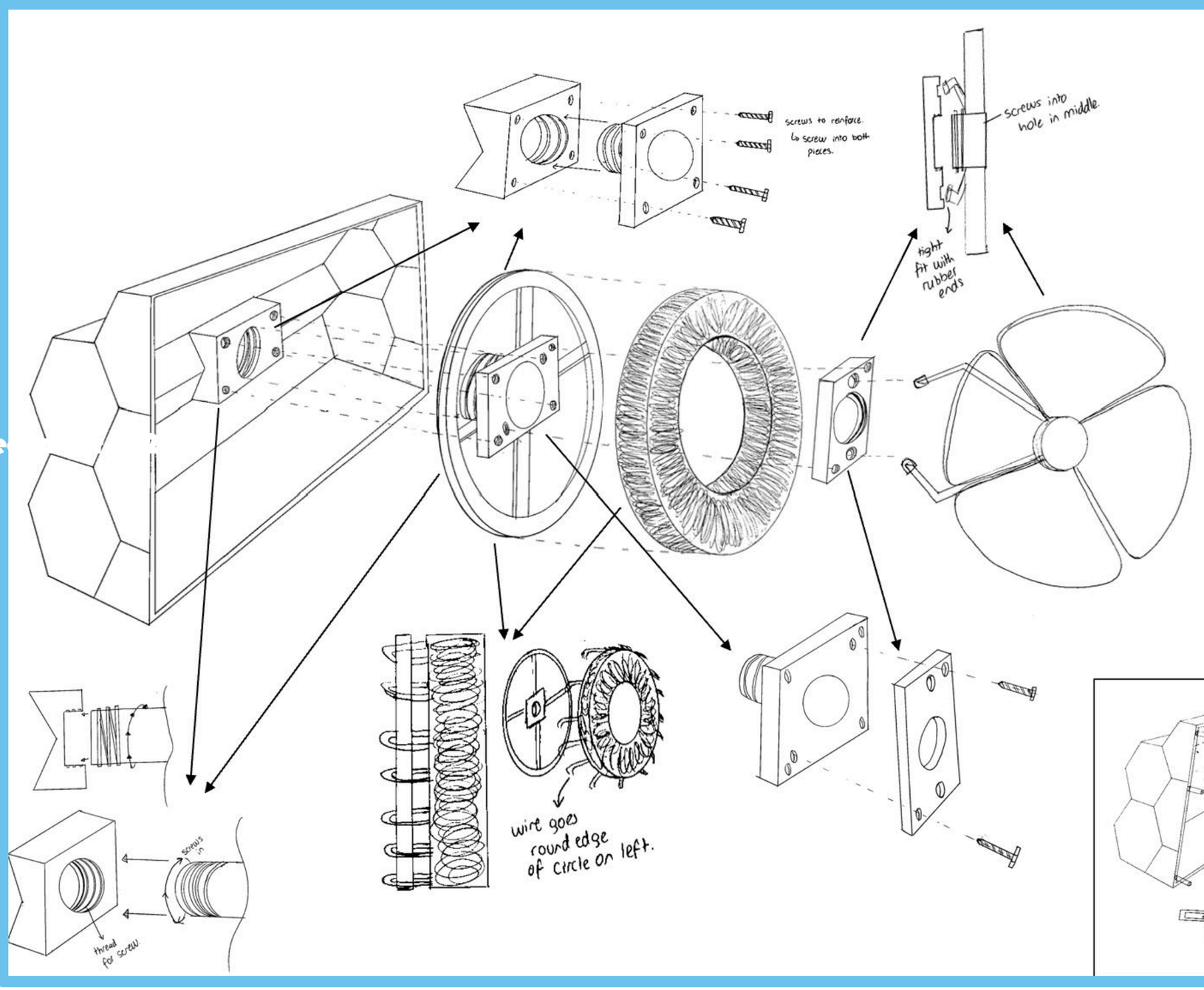


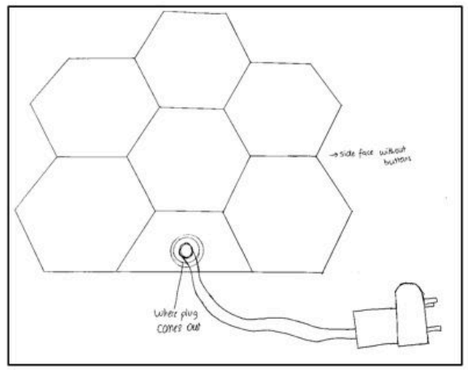
Concept refinement page two

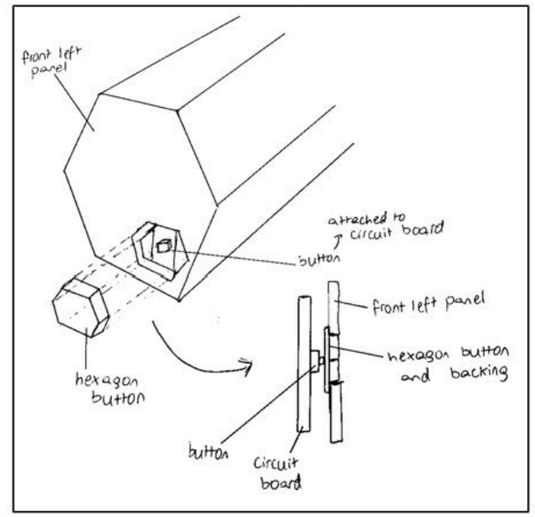


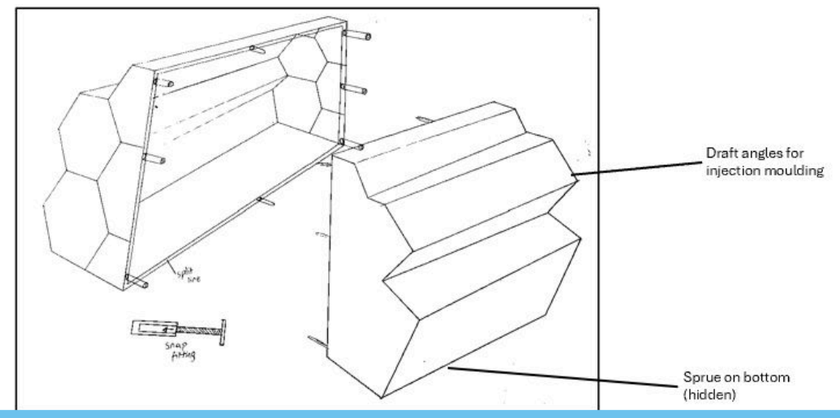
Final refinements

Project work: Heater project - exploded diagram



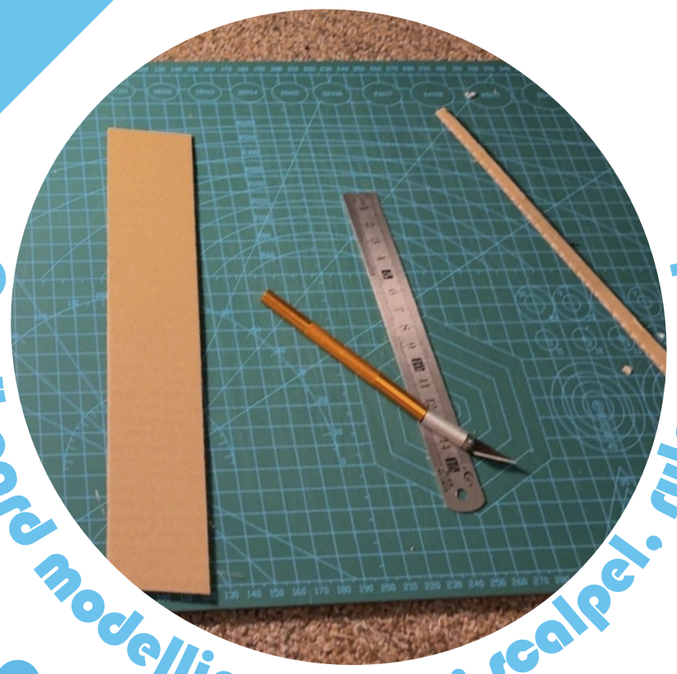




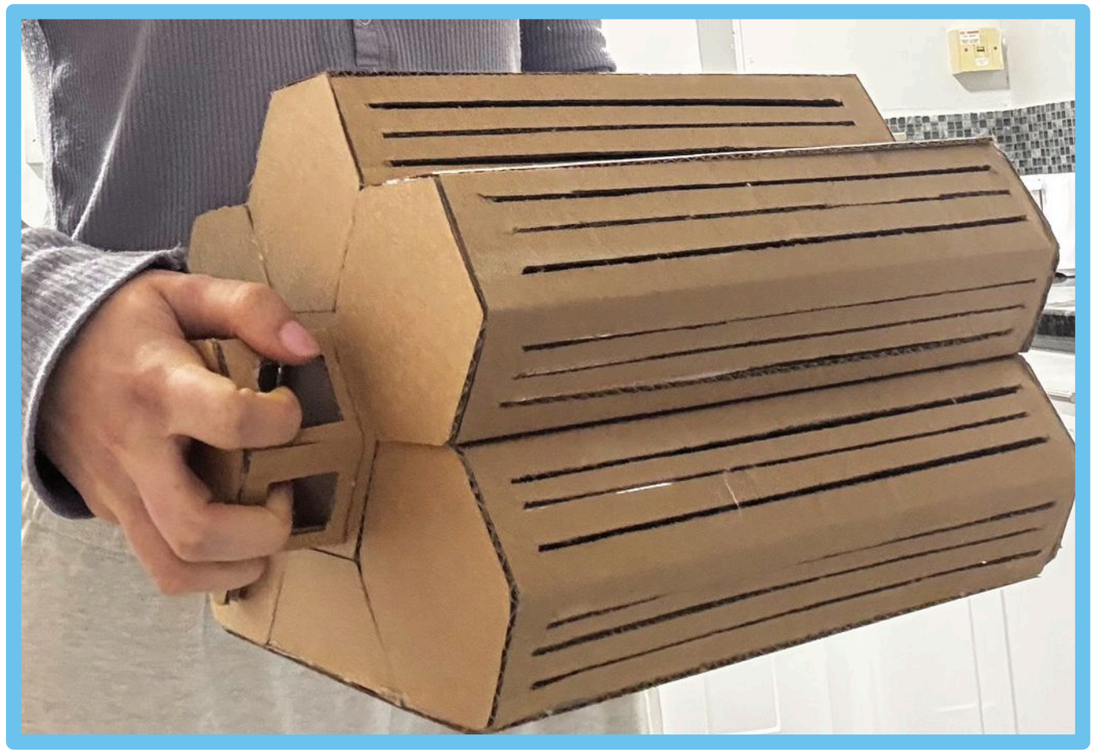


Exploded diagram showing inner workings of the heater and connection method of the parts. Used a heater that I took apart myself to see how the mechanics work.

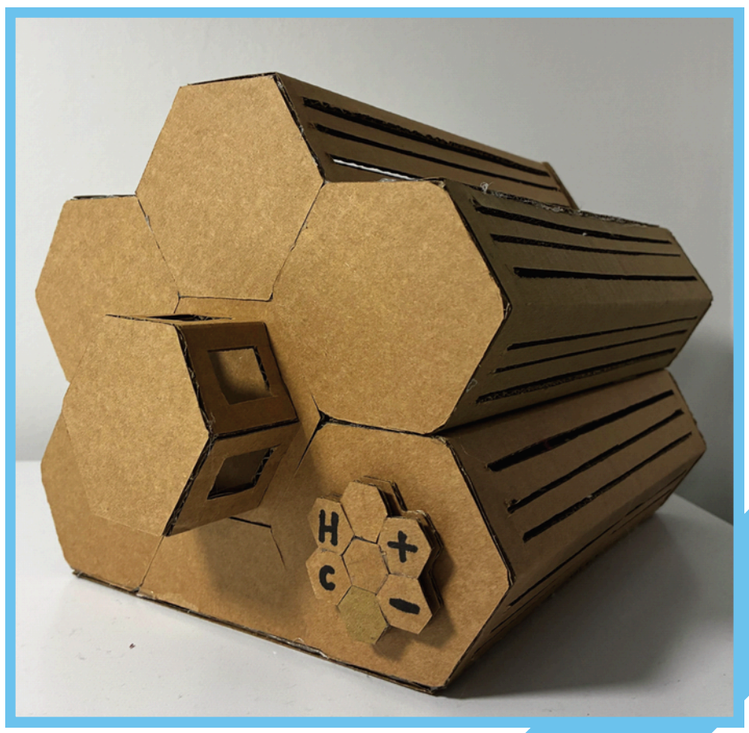
Project work: Heater project - cardboard modelling



Cardboard modelling set up: scalpel, ruler, mat

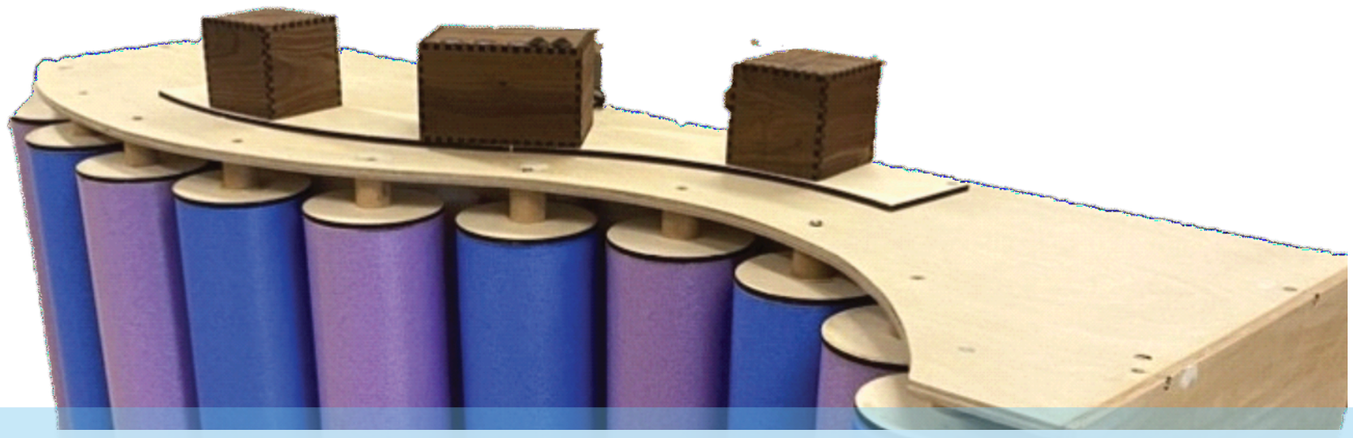


Final cardboard model of heater.
Shows ergonomic usage and detailed buttons.
Demonstrated to me aesthetics in shape and size which is hard to see through drawing.
I liked the soft edges of the product.





All pieces cut using template before gluing

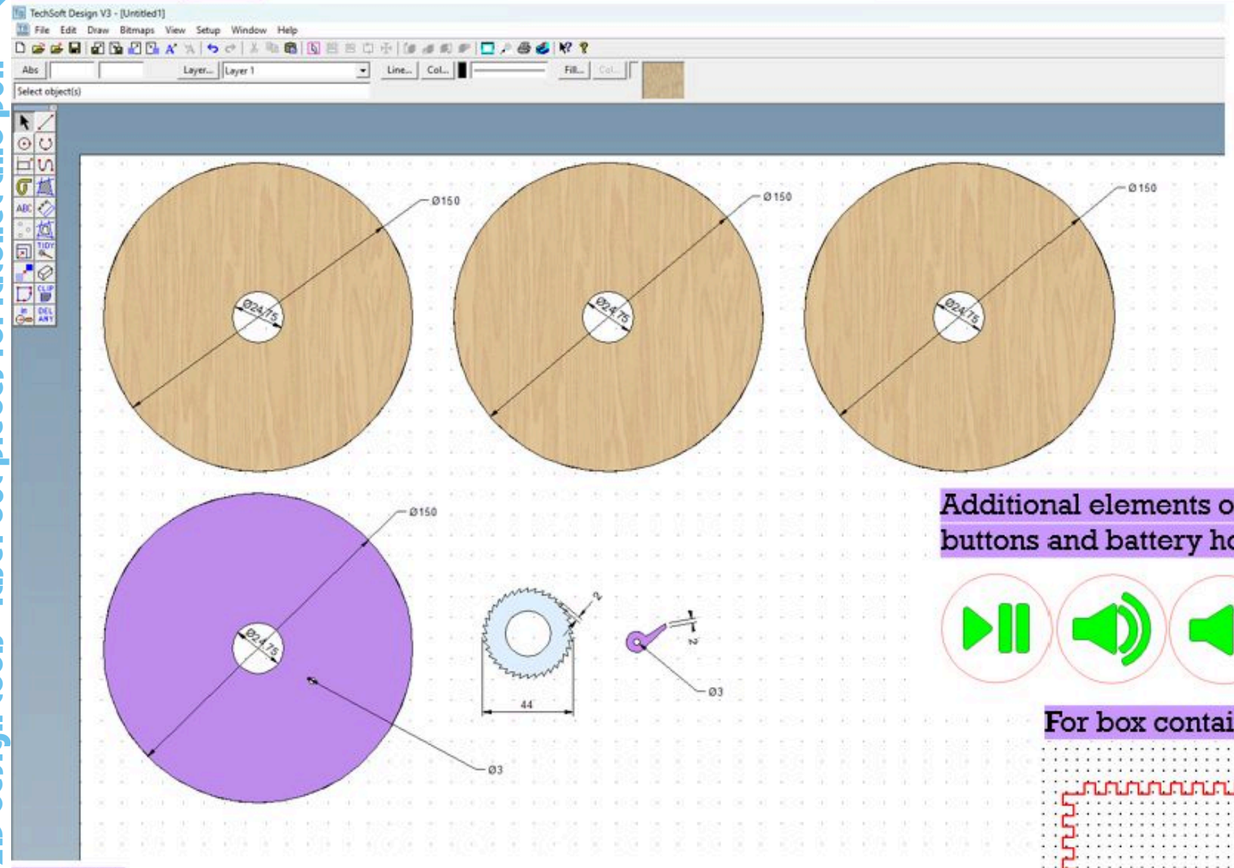


Sensory walkway

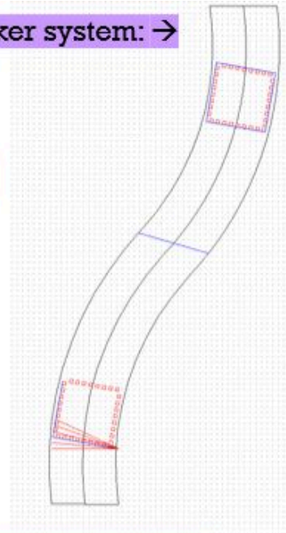
Project work: Sensory walkway - 2D Design Tools + SolidWorks

2D design tools - laser cut pieces for ratchet and pull

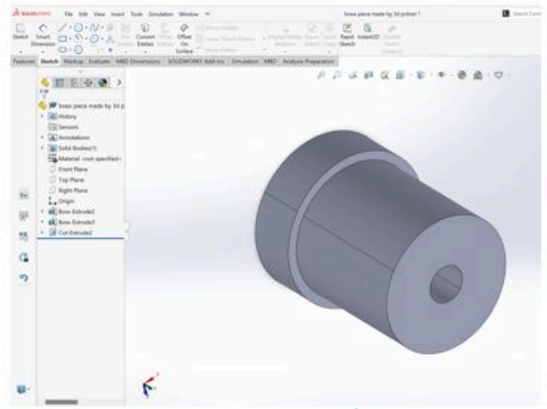
For ratchet system:



For platform holding speaker system: →

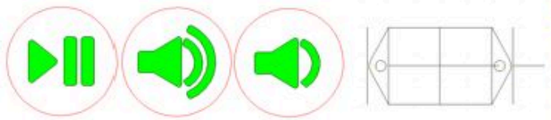


For spinning mechanism – to 3d print:

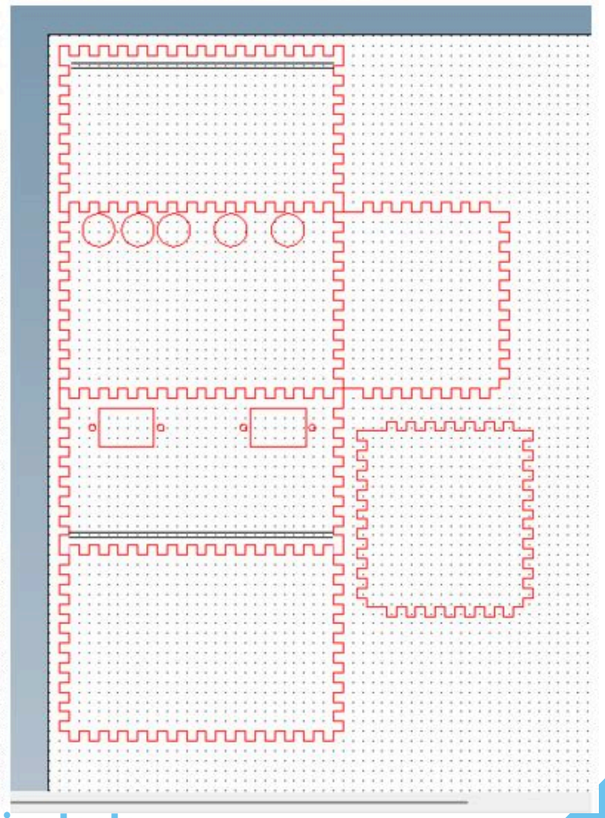


Solidworks component

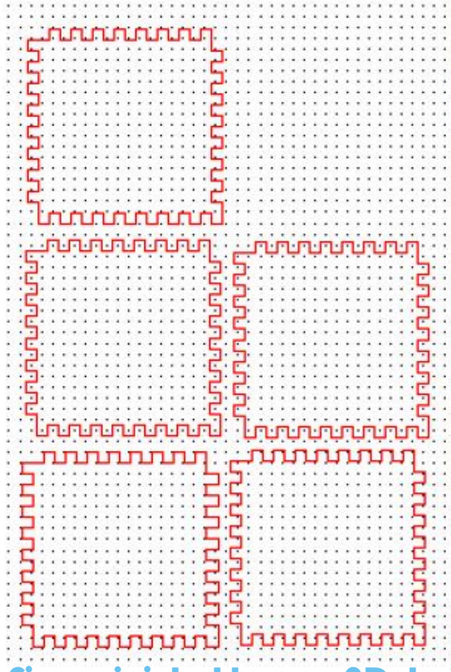
Additional elements of systems box – buttons and battery holder:



For box containing circuitry: :

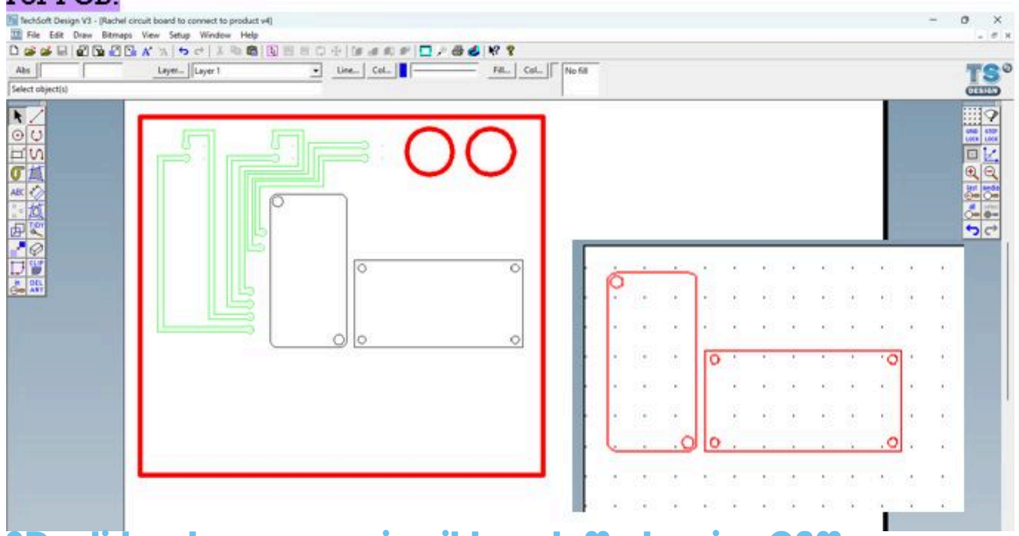


For box containing speakers:



Finger-jointed box on 2D design tools.

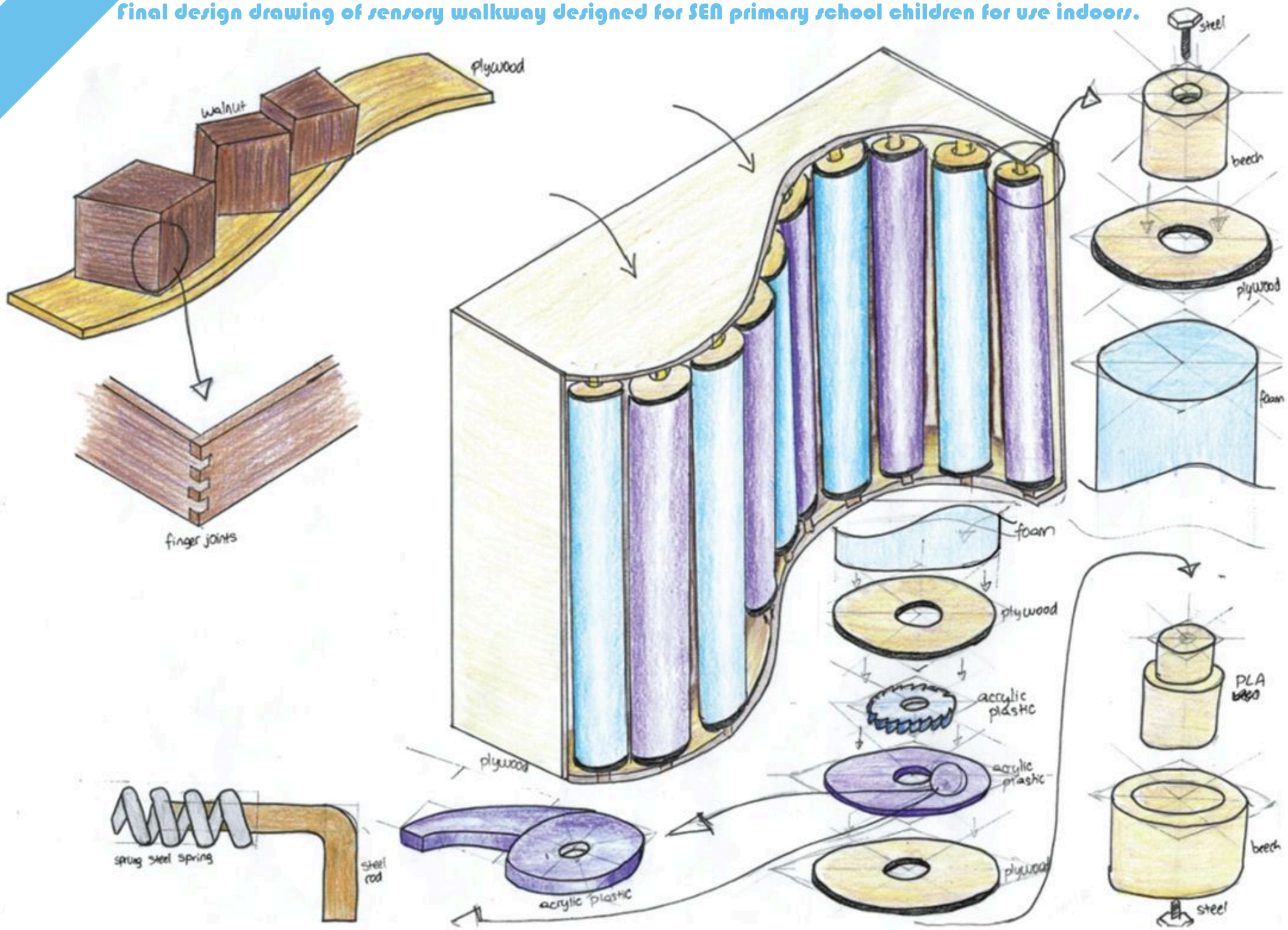
For PCB:



2D solidworks - copper circuit board. Made using CAM.

Project work: sensory walkway - final design drawing

Final design drawing of sensory walkway designed for SEN primary school children for use indoors.



Project work: Sensory walkway - user testing



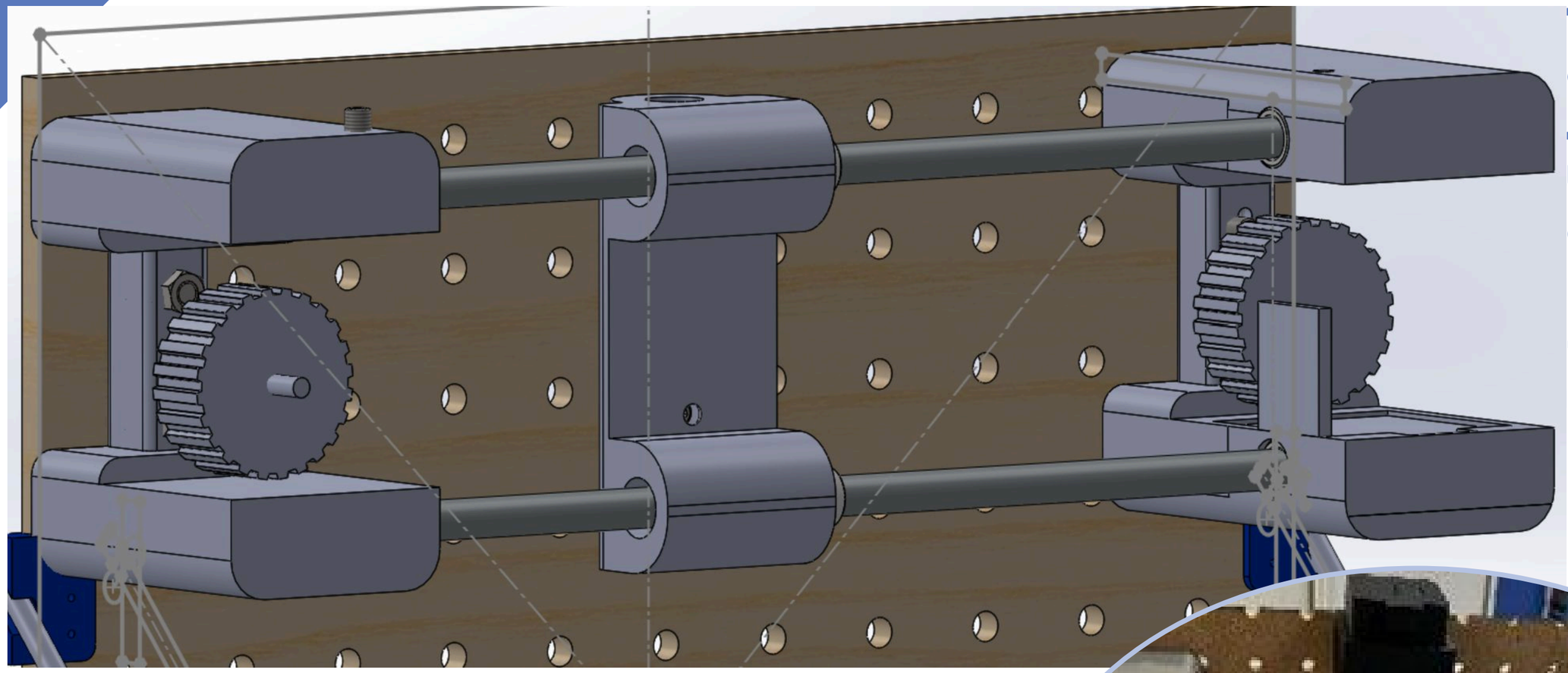




Use of my product by stakeholders. These are children of similar age to those who would use the product. I interviewed them numerous times throughout the design stage, and at the end to evaluate. The product was a success in that they enjoyed interacting with it.

Technical skills

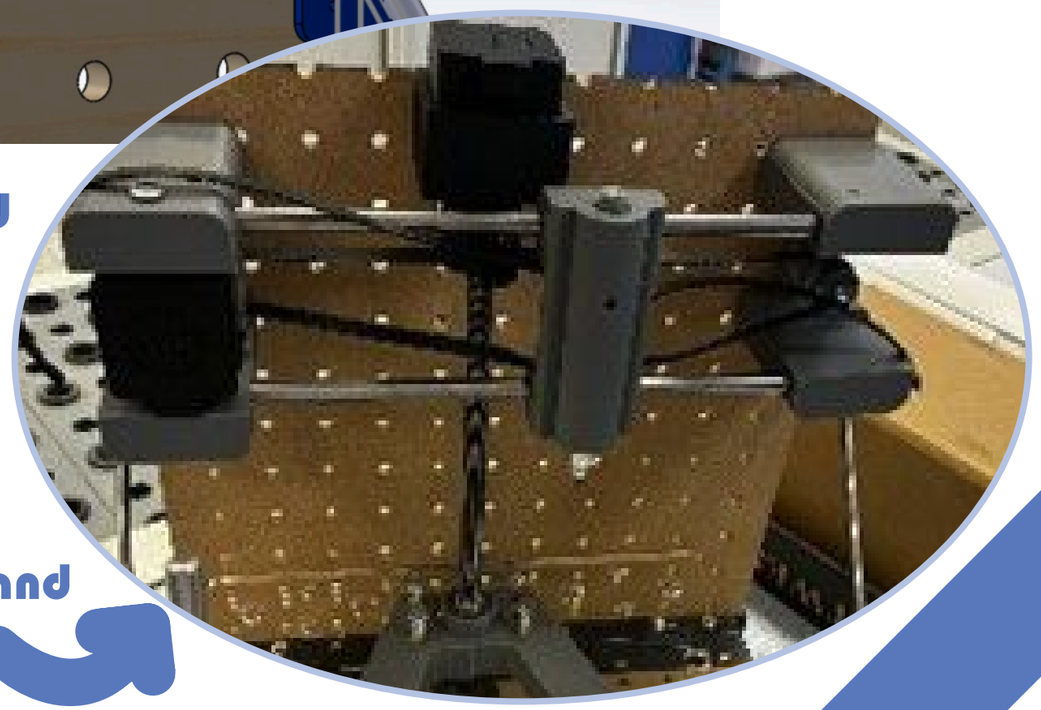
Technical skills: SolidWorks assembly

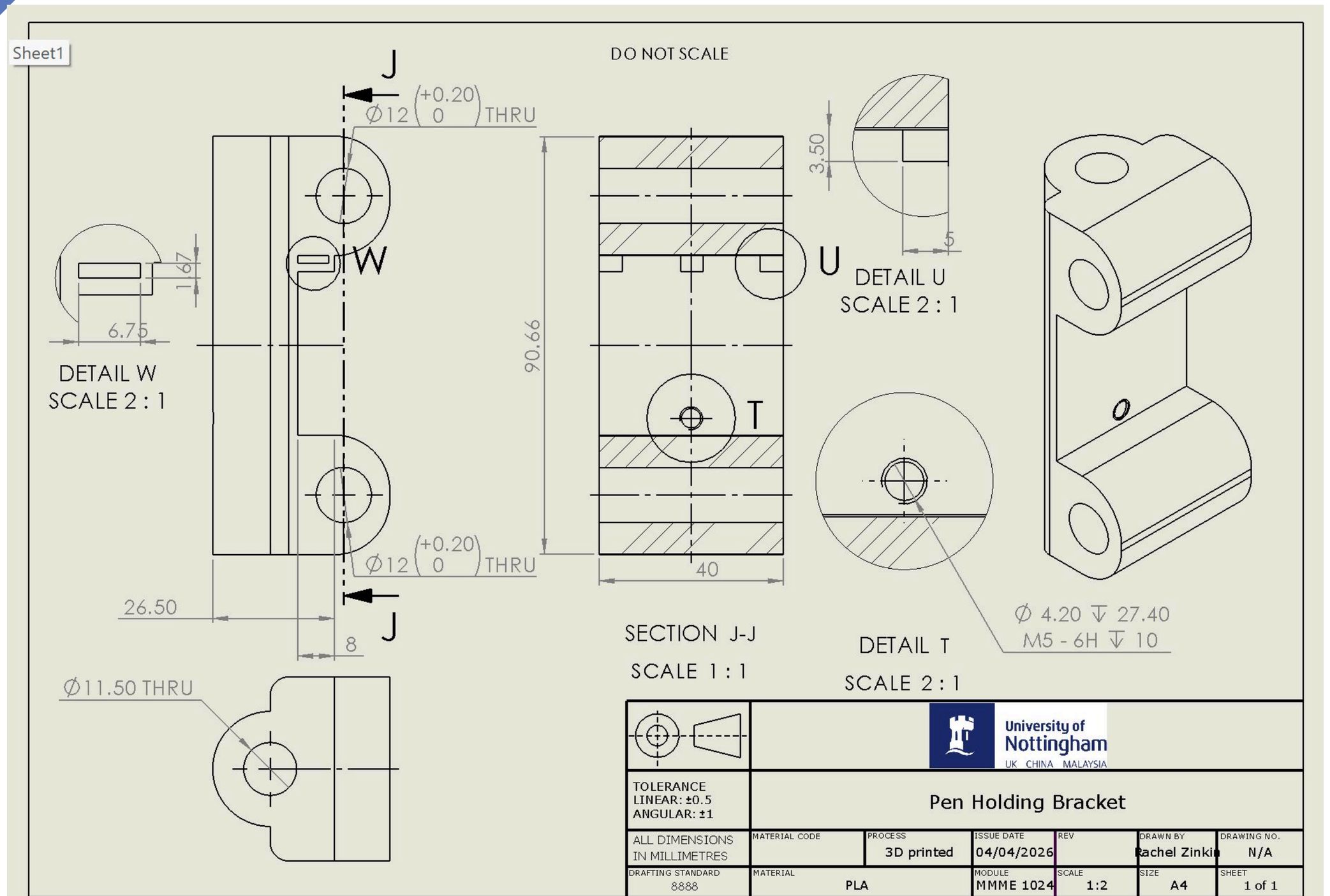


The parts were designed separately and then I used the assembly feature on SolidWorks to put it together.

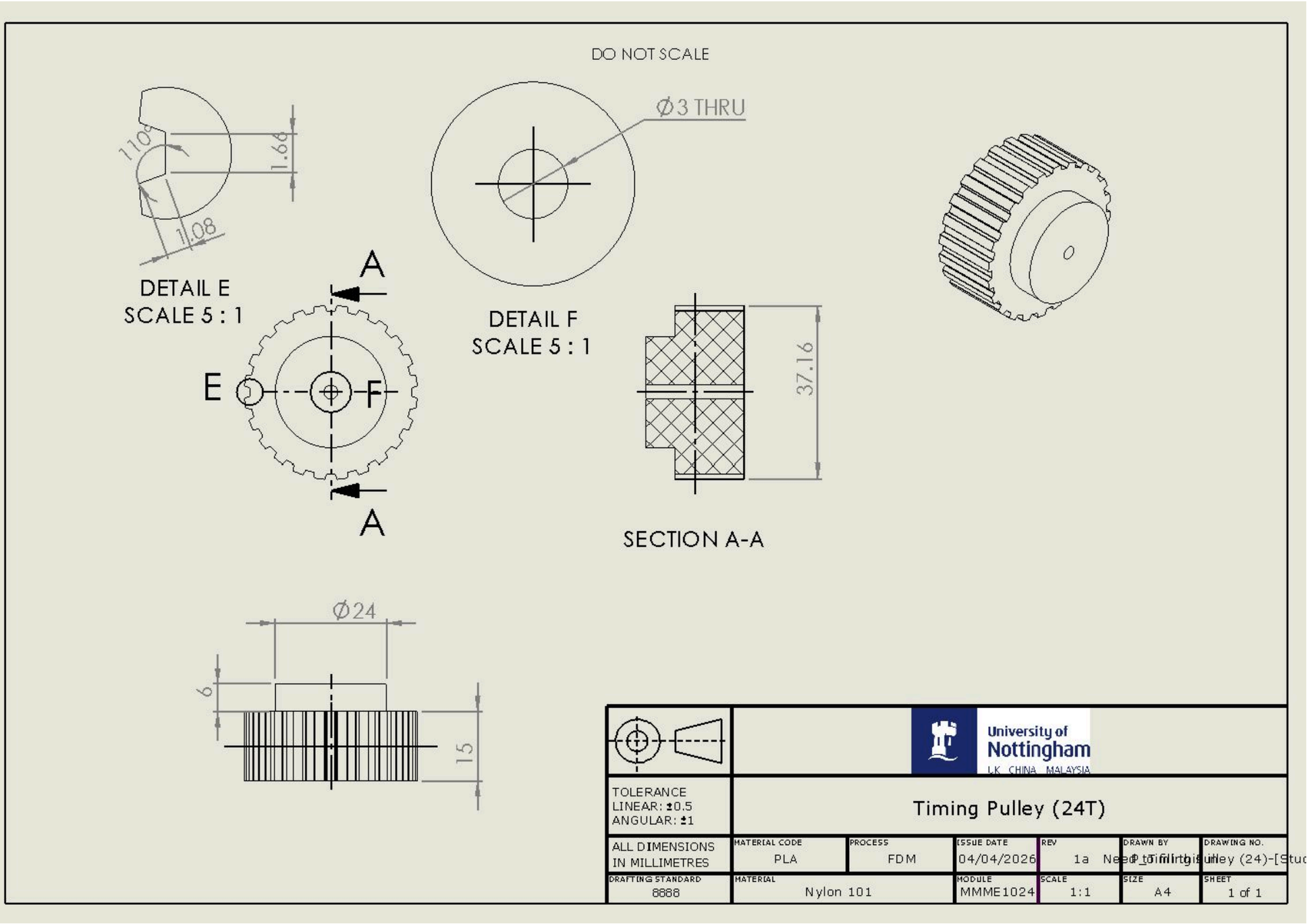
Belt and pulley system to move middle bracket left and right. I designed this using SolidWorks and some standard components provided to me. The three big brackets were all designed by me and my partner. The middle one holds a pen, and as it slides left and right a line is drawn.

The finished system is shown here complete with motors and belt.





Detailed drawing of 3D-printed component of above system.



Detailed drawing of 3D-printed standard component of above system.

Technical skills: Cardboard modelling





Practising cardboard modelling before prototyping my own designs.